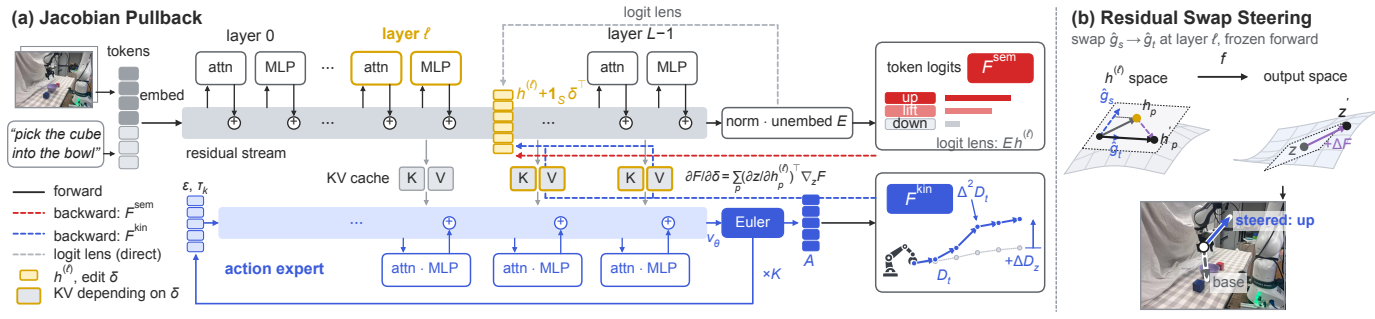

(a) Jacobian Pullback
tokens
embed
"pick the cube into the bowl"
layer 0
layer ℓ
logit lens
layer L−1
attn
MLP
residual stream
norm · unembed E
token logits
F^sem
up
lift
down
logit lens: Eh^(ℓ)
h^(ℓ)+1_S δ^⊤
∂F/∂δ = Σ_p (∂z/∂h_p^(ℓ))^⊤ ∇_z F
KV cache
K
V
ε, τ_k
action expert
attn · MLP
Euler
v_θ
×K
A
F^kin
Δ²D_t
D_t
+ΔD_z
forward
backward: F^sem
backward: F^kin
logit lens (direct)
h^(ℓ), edit δ
KV depending on δ
(b) Residual Swap Steering
swap ĝ_s → ĝ_t at layer ℓ, frozen forward
h^(ℓ) space
f
output space
ĝ_s
ĝ_t
h_p
h'_p
z
z'
+ΔF
steered: up
base